Speech Denoising on Noisy AudioMNIST with a U-Net

Spectrogram-to-spectrogram denoising of spoken digits: a network maps the 128 x 128 z-normalised log-mel spectrogram of a noisy recording to that of the clean recording. A baseline convolutional autoencoder is compared with a U-Net using an ImageNet-pretrained ResNet-34 encoder.

Test MSE: autoencoder 0.7347, U-Net 0.1485. Test SNR improvement: autoencoder +0.70 dB, U-Net +7.70 dB. SNR is measured on the normalised log-mel spectrograms, not on waveforms.

All figures, tables and numbers in this report are taken directly from the stored outputs of notebooks/speech_denoising_unet.ipynb.

Execution: complete original run on the full dataset (30,000 noisy/clean pairs; 24,000 / 3,000 / 3,000 train / validation / test) on a GPU: 10 epochs for the autoencoder, 5 epochs for the U-Net. This is not a partial run.

Convolutional autoencoder: test-set SNR on normalised log-mel spectrograms

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Average Noisy SNR: 1.80 dB
Average Denoised SNR: 2.50 dB
SNR Improvement: 0.70 dB
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U-Net (ResNet-34 encoder): test-set SNR on normalised log-mel spectrograms

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Average Noisy SNR: 1.80 dB
Average Denoised SNR: 9.49 dB
SNR Improvement: 7.70 dB
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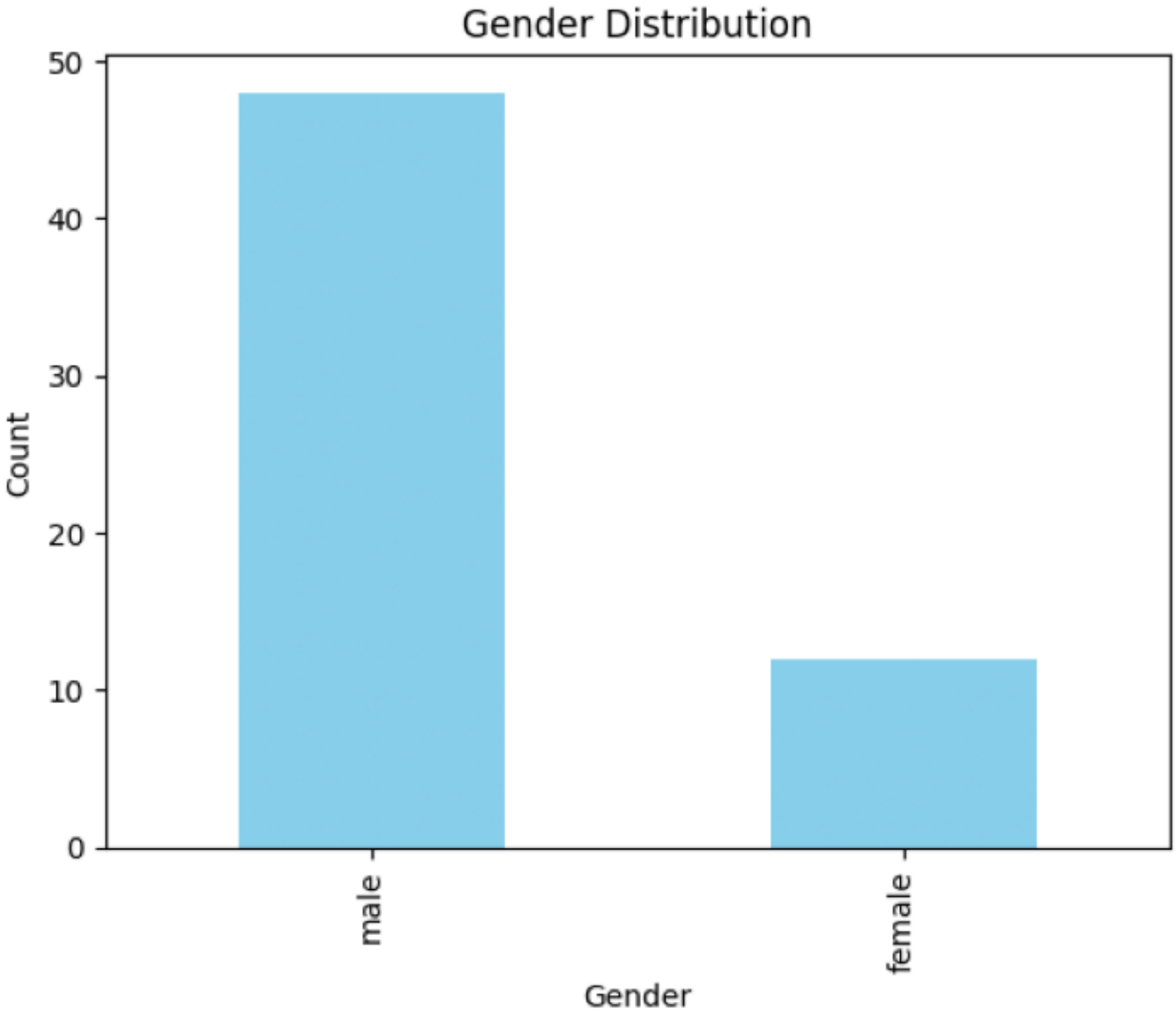


Figure 1. Speaker gender distribution.

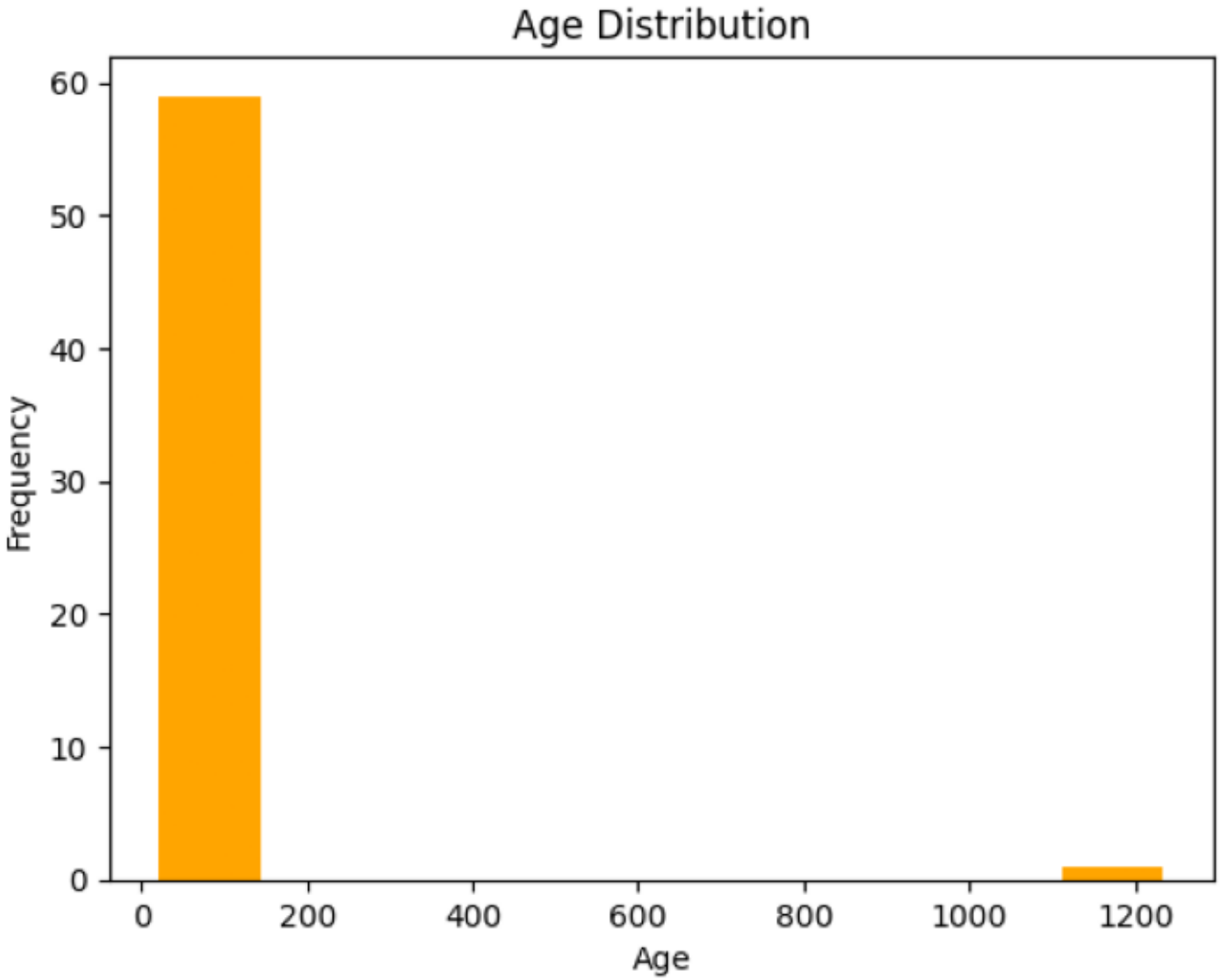


Figure 2. Speaker age distribution.

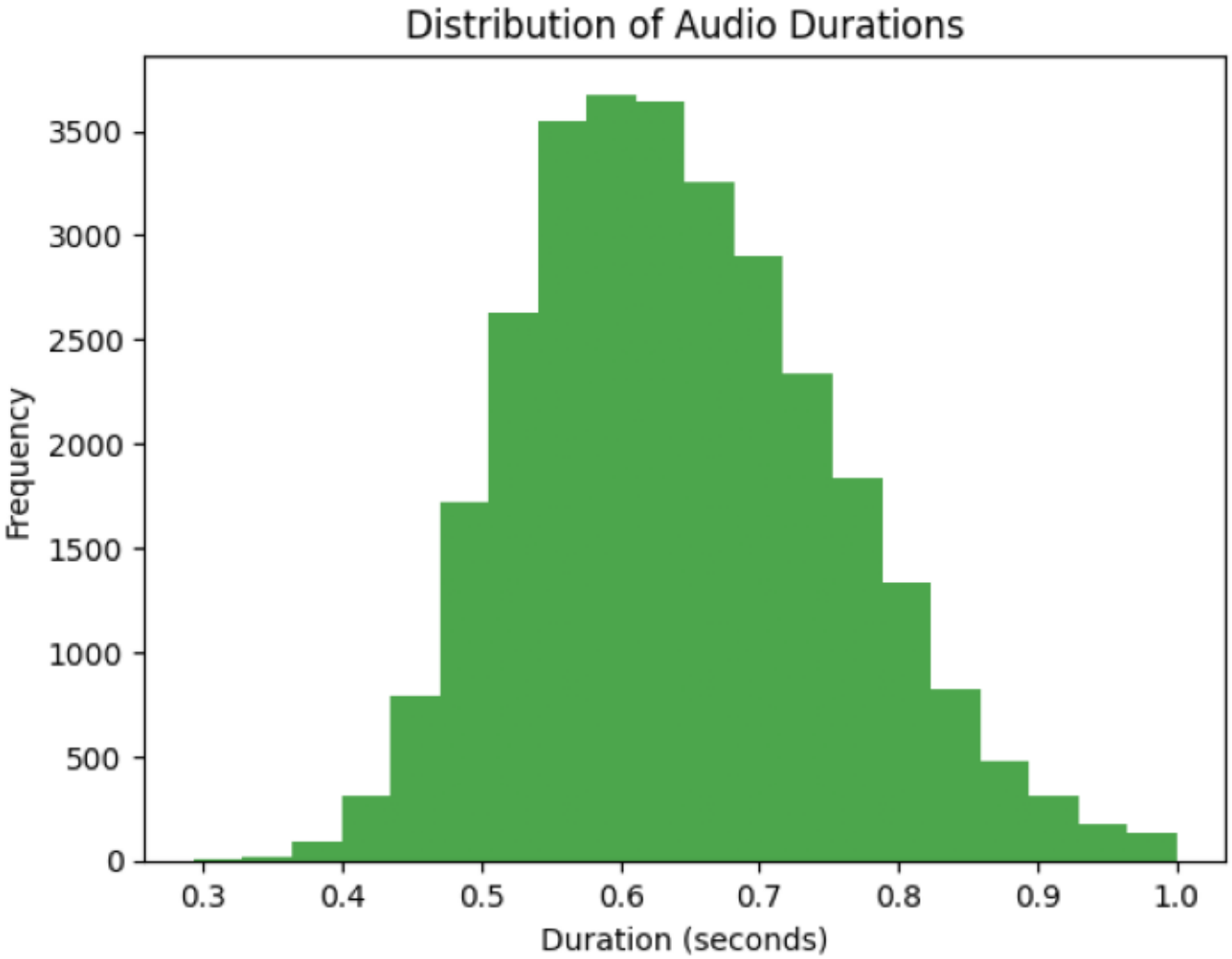


Figure 3. Distribution of clean-recording durations (seconds).

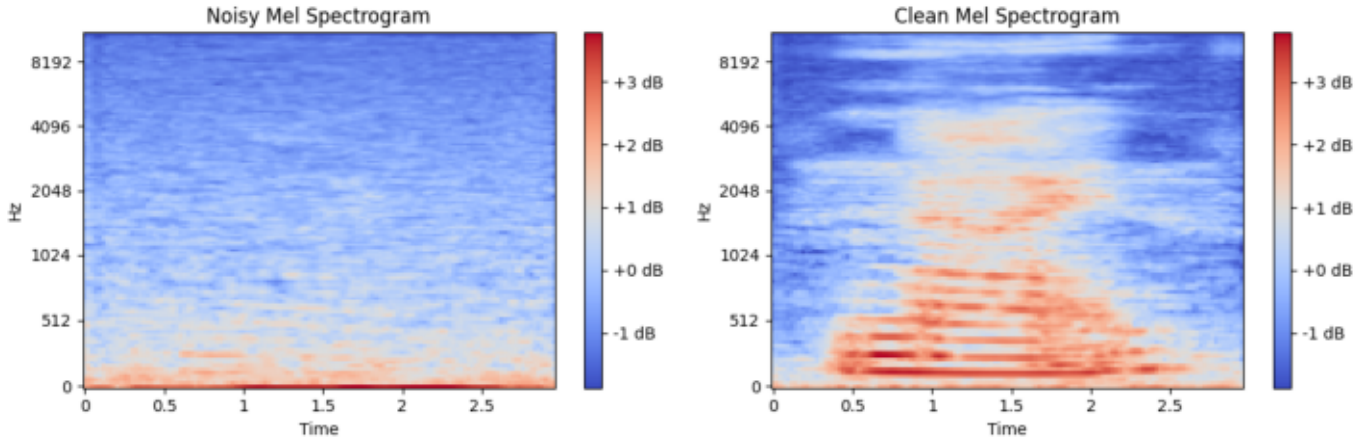


Figure 4. Log-mel spectrograms of one noisy/clean training pair (z-normalised values).

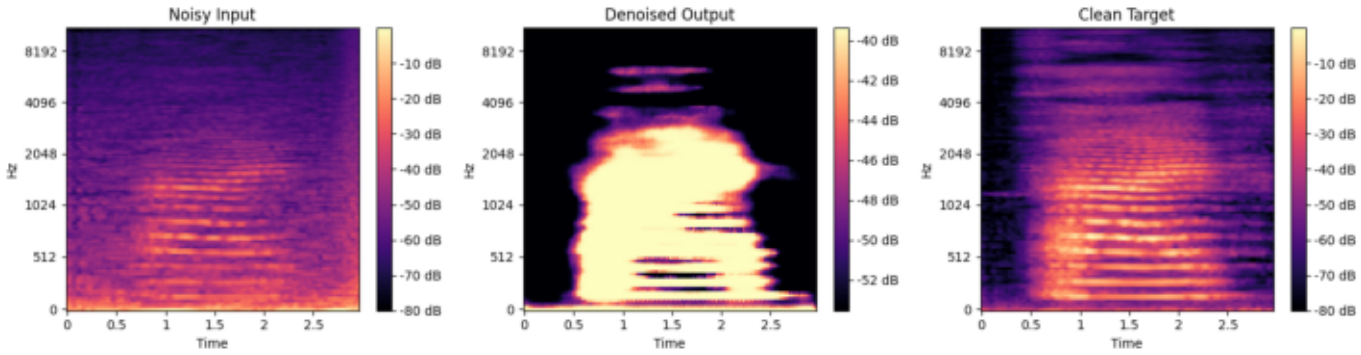


Figure 5. Autoencoder on a test sample: noisy input, denoised output, clean target.

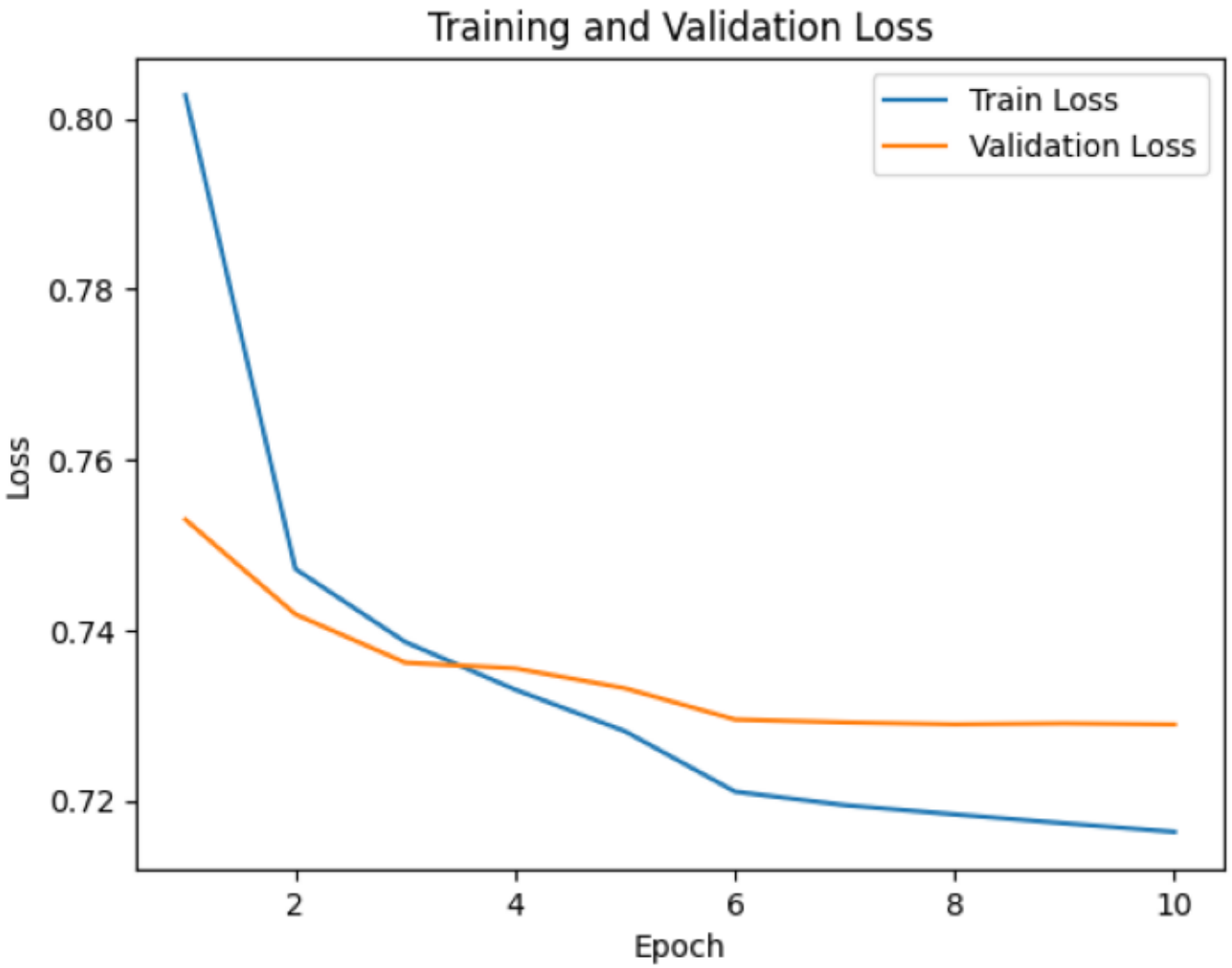


Figure 6. Autoencoder training and validation MSE.

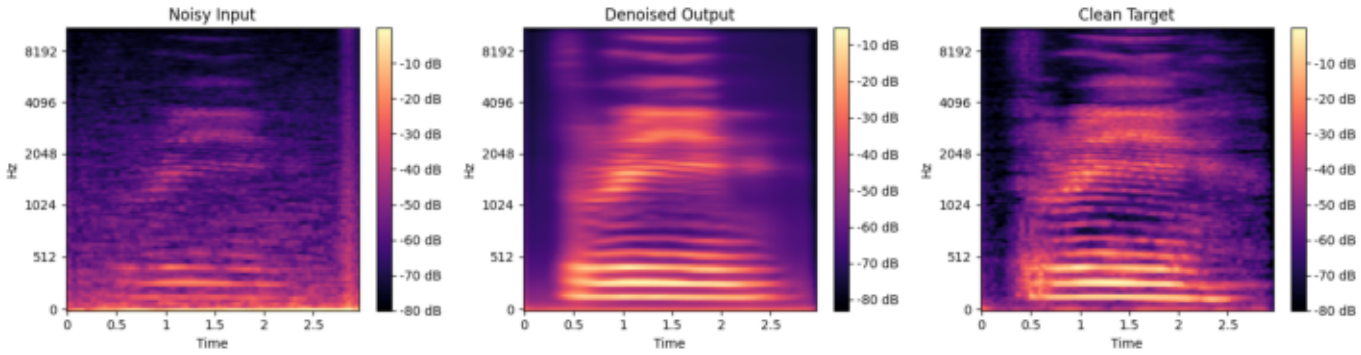


Figure 7. U-Net on a test sample: noisy input, denoised output, clean target.

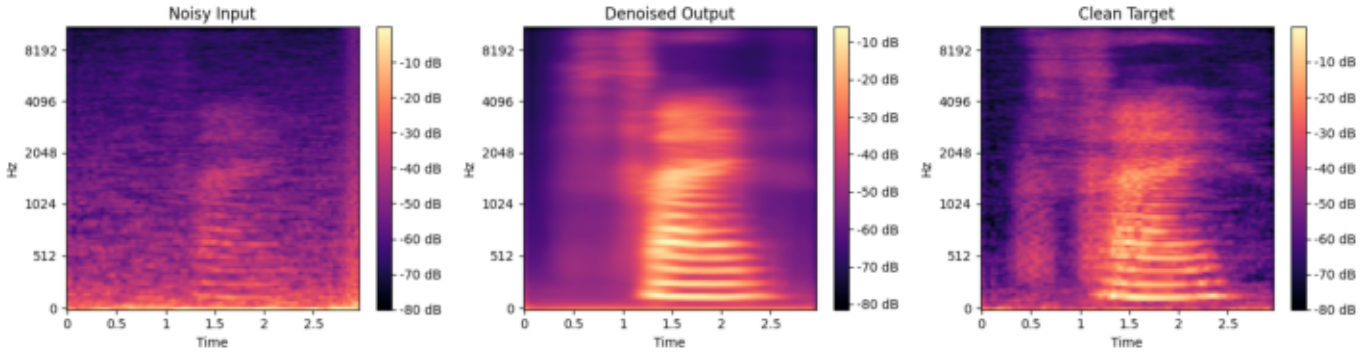


Figure 8. U-Net on a second test sample.

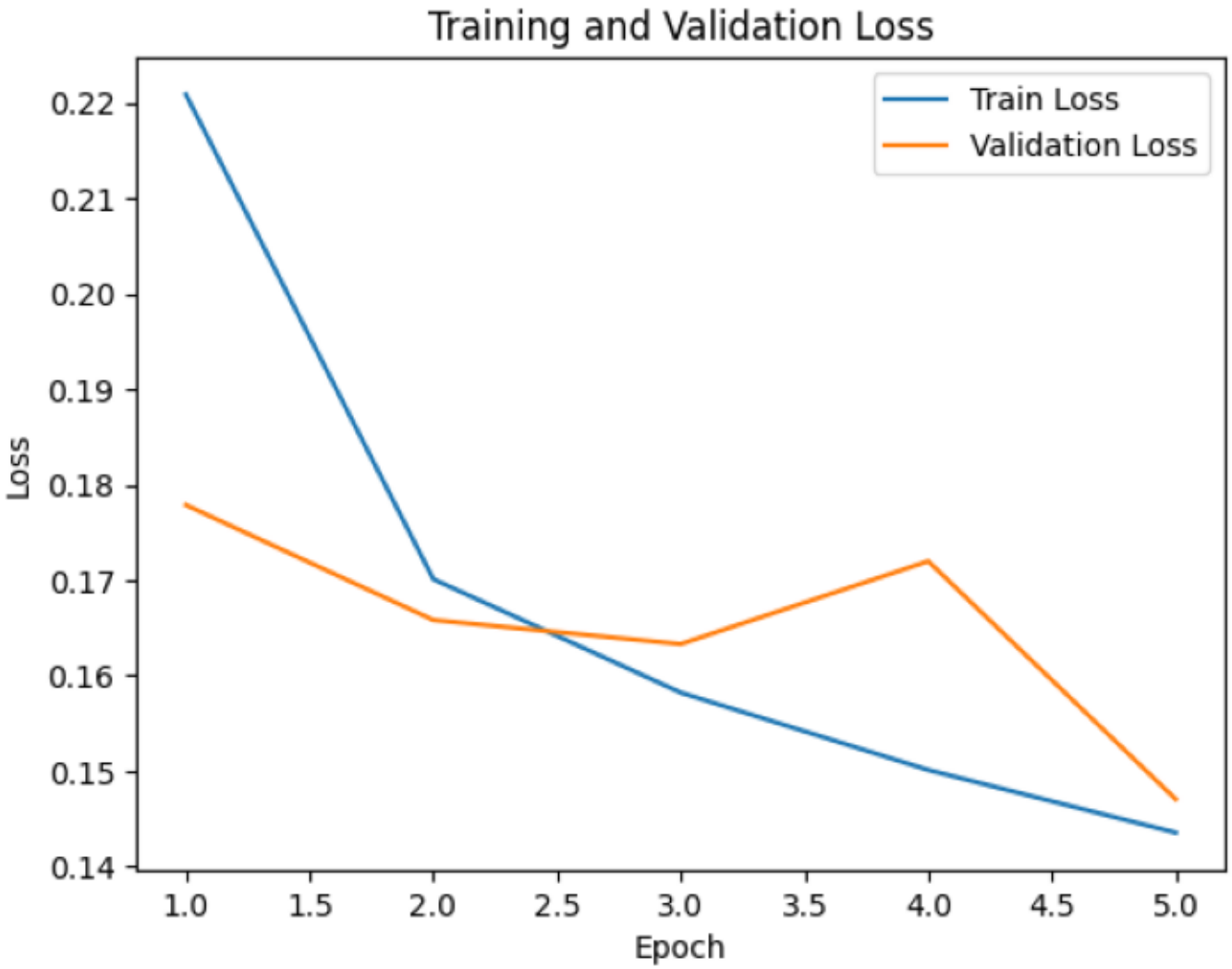


Figure 9. U-Net training and validation MSE.

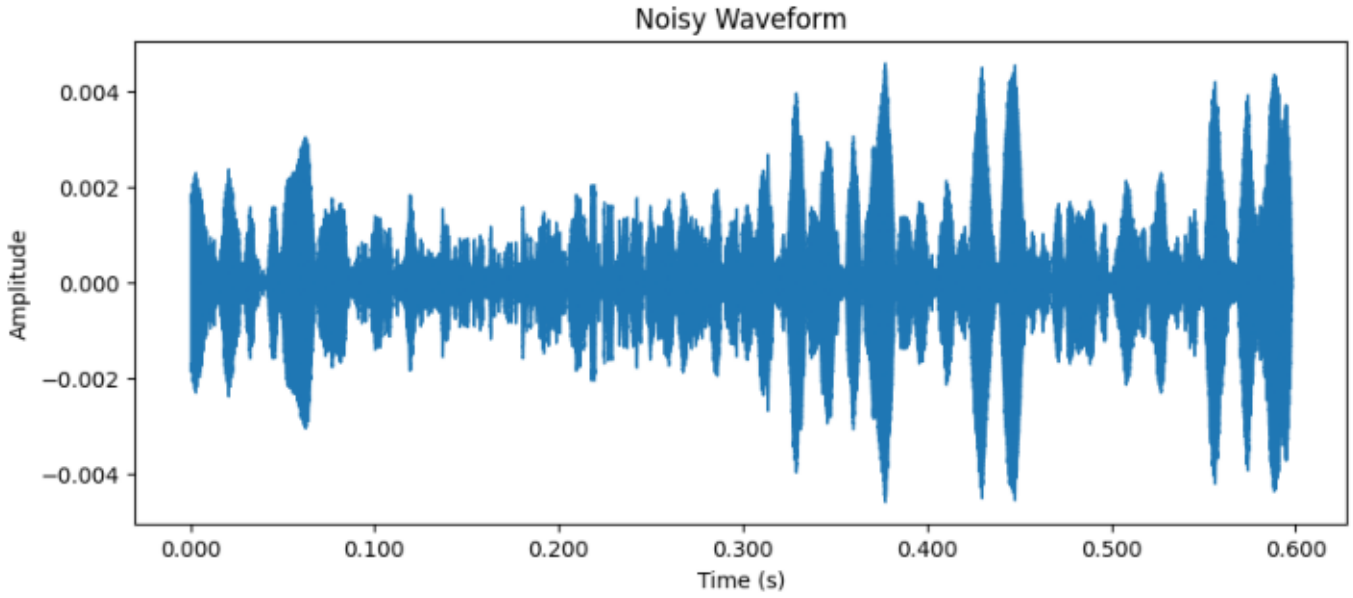


Figure 10. Waveform reconstructed with Griffin-Lim from the noisy spectrogram.

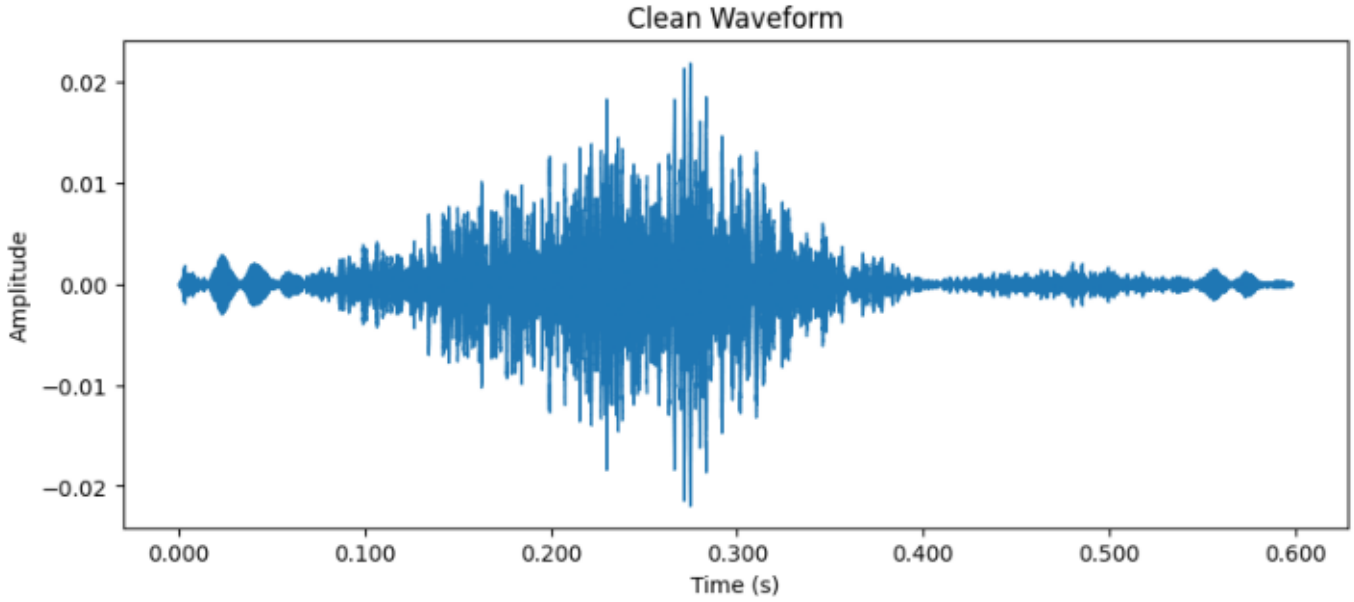


Figure 11. Waveform reconstructed with Griffin-Lim from the clean spectrogram.

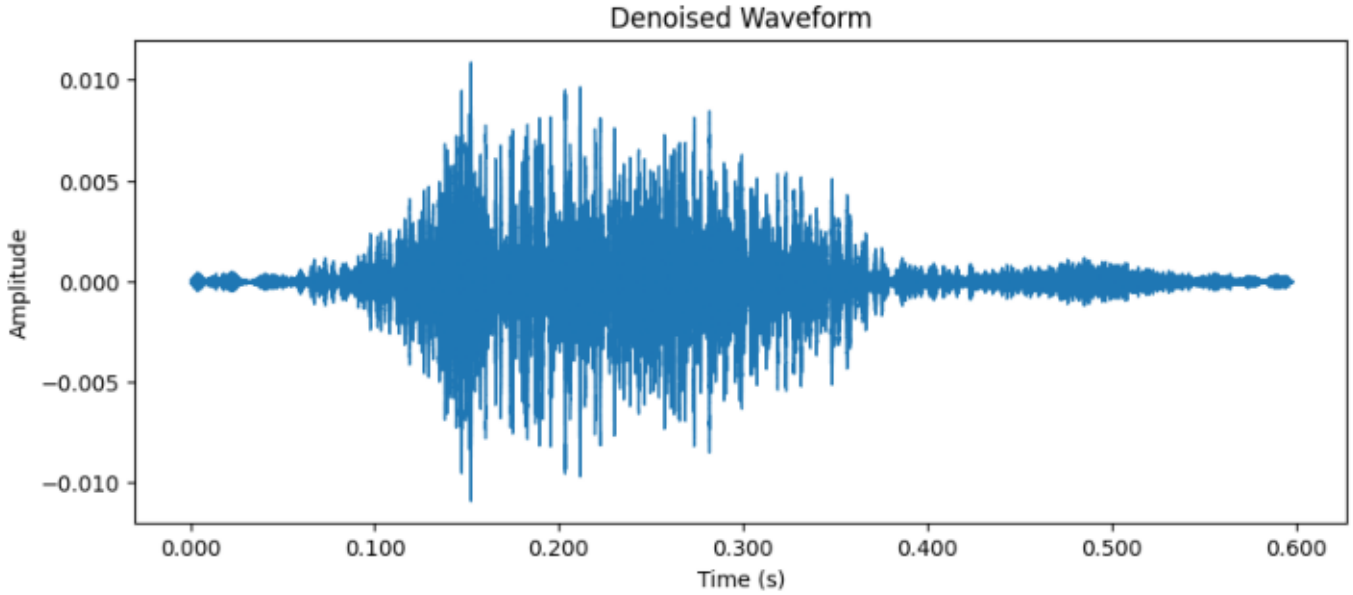


Figure 12. Waveform reconstructed with Griffin-Lim from the U-Net output.

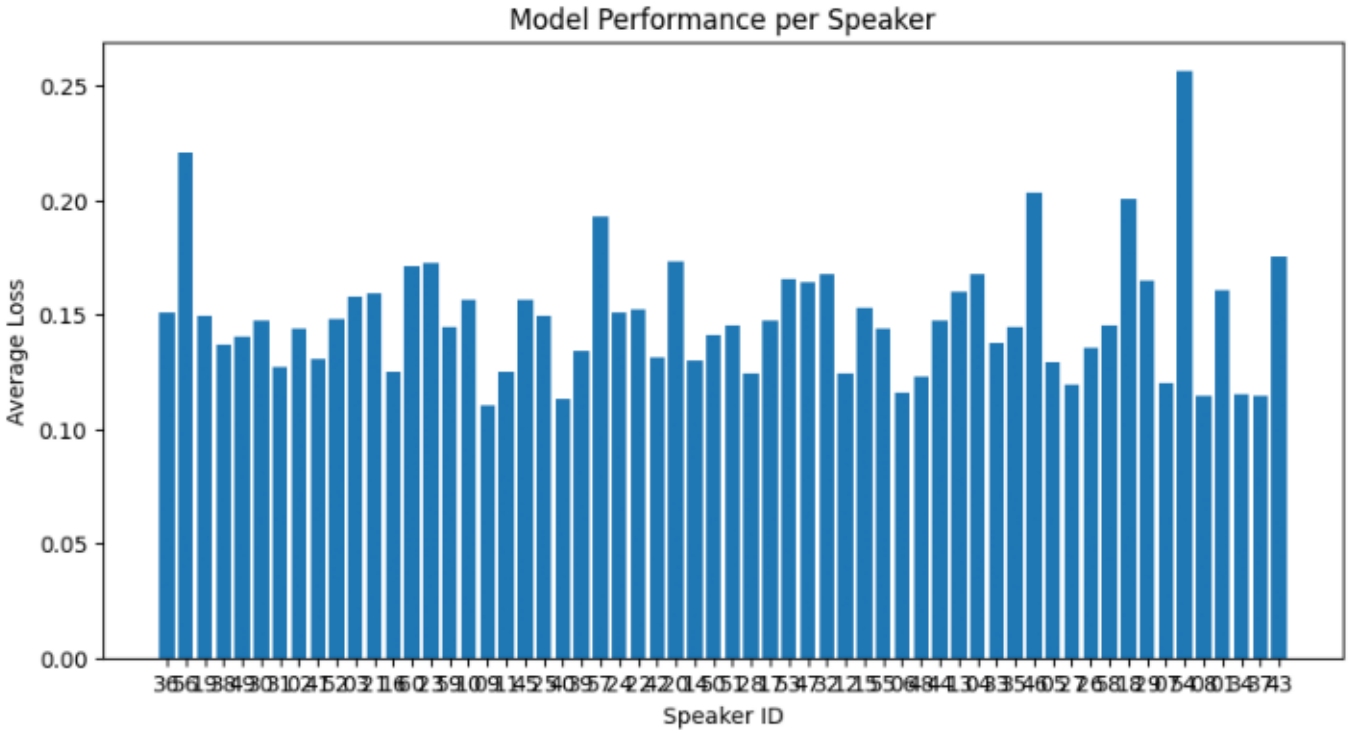


Figure 13. U-Net test MSE per speaker.

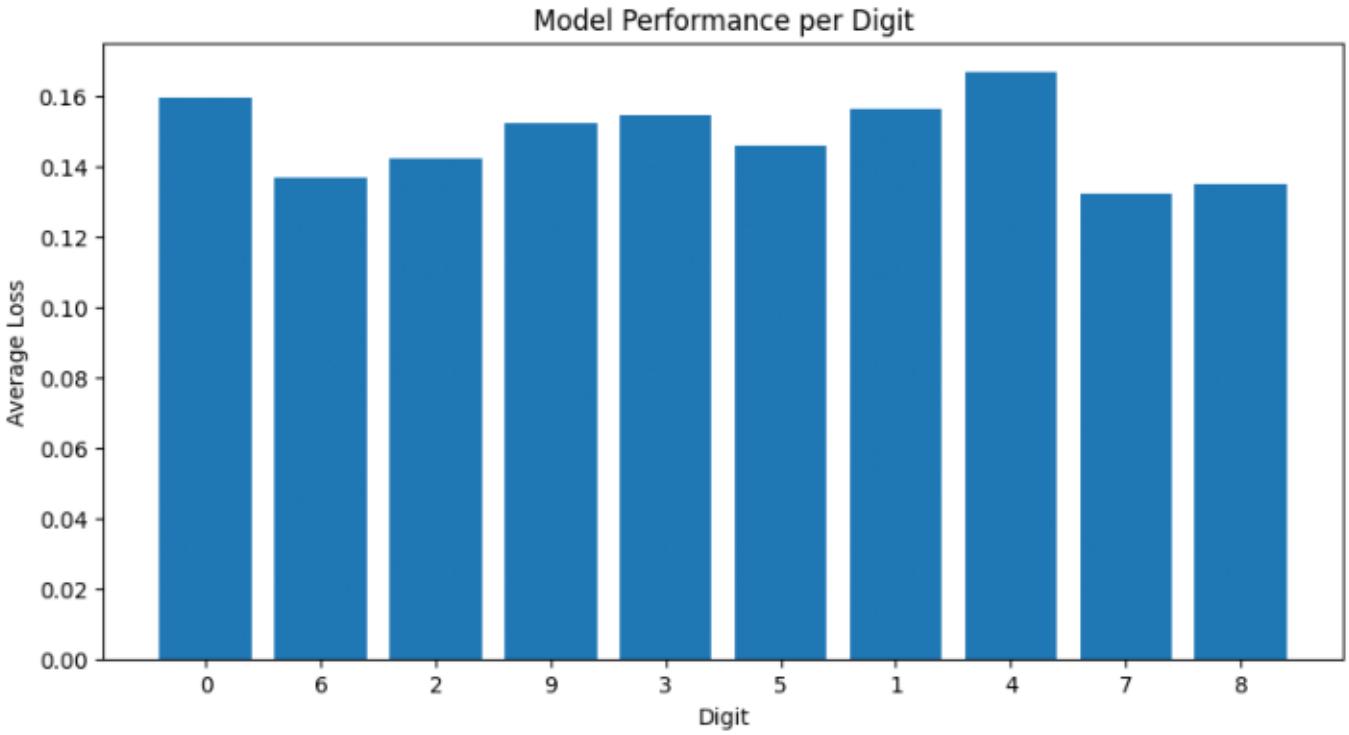


Figure 14. U-Net test MSE per spoken digit.